

Three Stage Drawing Transfer

Collaborative drawing between a Generative Adversarial Network, co-robotic arm, and five-year-old child

Robert Twomey

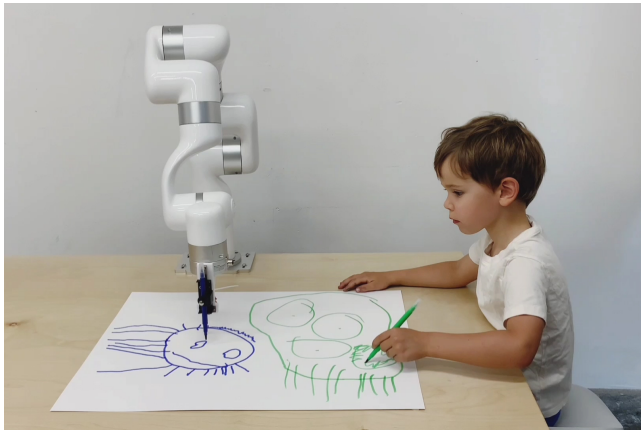
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1 PROJECT URL

<https://roberttwomey.com/three-stage-drawing-transfer/>



2 DESCRIPTION OF ARTWORK

Three Stage Drawing Transfer is an experimental human-robot performance and emerging media arts research project exploring drawing, cognition, and new possibilities for creative, embodied interactions between humans and machines. The work creates a visual-mental-physical circuit between a Generative Adversarial Network (GAN), a co-robotic arm, and a five-year-old child. Building on traditions from experimental performance and collaborative drawing, it extends these with the emergent capabilities of generative neural networks and robotic automation. The live child-robot interaction occurs within the space of a shared sheet of paper, where they together develop a collaborative drawing guided by the GAN's

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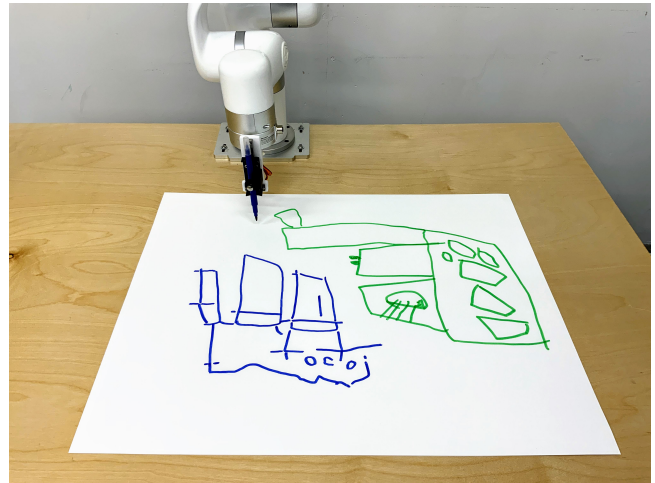
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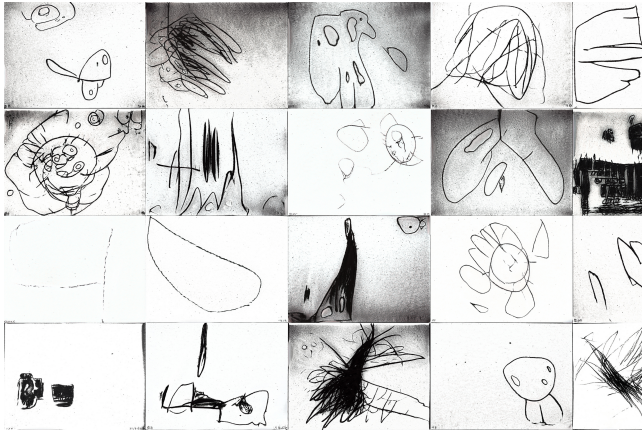
understanding and the child's imagination, resulting in a drawing as artifact and video as documentation.



The title refers to Dennis Oppenheim's 1971 performance 'Two Stage Transfer Drawing' [Oppenheim 1971], a direct inspiration for this piece. In that work, Oppenheim staged a photographic performance with his son, reconceiving drawing as a form of intimate, embodied, intergenerational exchange. Here, I have added additional translational stages between human and non-human participants: from the host of absent children whose artworks form the training data for a blackbox neural network; through the robotic arm manifesting the neural network's latent images with pen on paper; to my son, interpreting and augmenting these generative outputs with his own hand and imagination.

The neural network at the center of this artwork is a StyleGAN2 architecture [Karras et al. 2020] trained on a collection of over 7,000 children's drawings gathered by Rhoda Kellogg [Kellogg 1969], a prominent early childhood educator and scholar. As visual subject matter, these "early graphic expressions" recall Jean Dubuffet's art brut (raw art). I am interested in children's drawings as subject matter not only for being outside of taste, convention, and learned expression—as Dubuffet was—but also for how they show representation and visual language in their processes of development. Perhaps a better reference for this endeavor are the Surrealists, who also sought to escape the personal bounds of language and subjectivity through sleep deprivation, mind-altering substances, and collaborative and chance processes.

This project explores key dimensions of embodied interaction and generative art. It takes the outputs of a GAN, typically ephemeral



and represented as on-screen images or video, and concretizes them through the robotic drawing apparatus. It creates a performative encounter between the child and robot, emphasizing the child's active involvement with the machine as co-equal artists inhabiting a room. Finally, it juxtaposes the live perception, cognition, and creativity of the child (in interpreting and responding to the robot) with the "learning" and "creativity" of the GAN-plus-robot system.

These questions of where we search for the other; when we attribute autonomy and intelligence; and why we might wish to

escape our human subjectivities speak to core issues in the design and use of robotics and AI. This project is one attempt to think through those questions in an embodied way.

3 ARTIST INFORMATION

Robert Twomey is an artist and engineer exploring poetic intersections of human and machine perception, particularly how emerging technologies transform sites of intimate life. He has presented his work at SIGGRAPH (Best Paper Award), CVPR, ISEA, NeurIPS, the Museum of Contemporary Art San Diego, and has been supported by the National Science Foundation, the California Arts Council, Microsoft, Amazon, HP, and NVIDIA. He is an Assistant Professor of Emerging Media Arts with the Johnny Carson Center, University of Nebraska-Lincoln, and an Artist in Residence with the Arthur C. Clarke Center for Human Imagination, UC San Diego.

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